topic_3_crtp_pattern

Theory, Edge Cases & Code Examples

- ✓ CRTP Pattern Overview
- ✓ CRTP Benefits in Autonomous Vehicles
- ✓ CRTP vs Virtual Polymorphism Comparison
- ✓ Name Hiding in Template Base Classes
- ✓ Infinite Recursion Without Implementation Override

→ *Swipe to learn* →

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Theory

📌 1. CRTP Pattern Overview (1/2)

Definition: Curiously Recurring Template Pattern - a class inherits from a template base instantiated with the derived class itself.

Core Structure:

```
// CRTP Base Template
template <typename Derived>
class Base {
public:
    void interface() {
        // Static downcast to derived class (compile-time)
        static_cast<Derived*>(this)->implementation();
    }
};

// Derived class inherits from Base<Derived>
class Derived : public Base<Derived> {
public:
    void implementation() {
        std::cout << "Derived implementation\n";
    }
};

// Usage - polymorphic behavior without virtual functions
Derived d;
d.interface();  // Calls Derived::implementation() via static_cast
```

The "Curious Recursion":

Component	Role	Timing
`Base<Derived>`	Template base class	Instantiated per derived type
`Derived : public Base<Derived>`	Inheritance relationship	Derived passes itself as template arg
`static_cast<Derived*>(this)`	Downcast mechanism	Compile-time type conversion
Method dispatch	Polymorphism	Resolved at compile-time (inlined)

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Theory

📌 1. CRTP Pattern Overview (2/2)

Key Mechanism:

```
// What happens during compilation:
template <typename Derived>
void Base<Derived>::interface() {
    static_cast<Derived*>(this)->implementation();
    // Compiler knows exact type of Derived at this point
    // No vtable lookup needed - direct function call
}

// For class Derived : public Base<Derived>
// Compiler generates:
void Base<Derived>::interface() {
    static_cast<Derived*>(this)->implementation();
    // Resolves to: this->Derived::implementation()
    // Can be fully inlined
}
```

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Theory

📌 2. CRTP Benefits in Autonomous Vehicles (1/3)

Performance-Critical Use Cases:

Use Case	Traditional (Virtual)	CRTP (Static)	Performance Gain
Sensor fusion loop (1kHz)	Virtual dispatch per call	Inlined static dispatch	~15-20% faster
Control algorithms (10kHz)	Vtable overhead	Zero overhead	~10-30% faster
Path planning evaluation	Runtime polymorphism	Compile-time selection	~5-15% faster
Kalman filter variants	Virtual update() calls	Templated policy	Fully optimized

Real-World Applications:

A. Zero-Overhead Sensor Processing:

```
template <typename SensorType>
class SensorProcessor {
public:
    void processFrame() {
        auto& sensor = static_cast<SensorType&>(*this);

        auto rawData = sensor.readRawData();        // Virtual: ~5ns overhead
        auto filtered = sensor.applyFilter(rawData); // CRTP: 0ns overhead
        sensor.publishData(filtered);
    }
};

class LidarSensor : public SensorProcessor<LidarSensor> {
    // Implementation inlined at compile-time
};
```

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Theory

📌 2. CRTP Benefits in Autonomous Vehicles (2/3)

B. Policy-Based Filter Selection:

```
template <typename FilterPolicy>
class StateEstimator : public FilterPolicy {
    // Use either KalmanFilter or ExtendedKalmanFilter
    // Decision made at compile-time
};

StateEstimator<KalmanFilter> linearEstimator;      // For linear systems
StateEstimator<ExtendedKalmanFilter> nonlinear;    // For nonlinear systems
```

C. Mixin Composition:

```
template <typename Derived>
class WithLogging {
    void log(const std::string& msg) {
        std::cout << typeid(Derived).name() << ": " << msg << "\n";
    }
};

template <typename Derived>
class WithCaching {
    std::unordered_map<int, Result> cache;
};

// Combine behaviors at compile-time
class SensorNode : public WithLogging<SensorNode>,
                   public WithCaching<SensorNode> {
    // Gets logging + caching with zero runtime overhead
};
```

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Theory

📌 2. CRTP Benefits in Autonomous Vehicles (3/3)

Key Advantages:

Advantage	Explanation	Autonomous Vehicle Benefit
Zero overhead	No vtable, fully inlined	Critical for real-time control loops
Compile-time safety	Interface errors caught at compile-time	Prevents runtime failures in production
Code reuse	Share implementations without virtual cost	Sensor drivers, filters, controllers
Policy selection	Choose algorithms at compile-time	Different filter types per vehicle config
Composability	Multiple inheritance without conflicts	Mix logging, metrics, caching behaviors

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Theory

📌 3. CRTP vs Virtual Polymorphism Comparison (1/4)

Dispatch Mechanism:

Aspect	CRTP (Static Polymorphism)	Virtual Functions (Dynamic Polymorphism)
Resolution	Compile-time	Runtime
Mechanism	Template instantiation + static_cast	Vtable lookup + virtual dispatch
Overhead	✅ Zero (inlined)	❌ 5-15ns per call
Code size	⚠️ Larger (per-type instantiation)	✅ Smaller (one vtable)
Type known	✅ At compile-time	❌ At runtime
Optimization	✅ Full inlining possible	⚠️ Limited (devirtualization rare)

📌 3. CRTP vs Virtual Polymorphism Comparison (2/4)

Performance Comparison (1M calls):

```
// Benchmark results:

// Virtual function call: ~50ms
class Base {
    virtual void process() = 0;
};
for (int i = 0; i < 1000000; ++i) {
    base->process();  // Virtual dispatch
}

// CRTP call: ~15ms
template <typename T>
class Base {
    void process() { static_cast<T*>(this)->impl(); }
};
for (int i = 0; i < 1000000; ++i) {
    derived.process();  // Inlined static call
}

// Speedup: ~3.3x faster
```

When to Use Each:

Use CRTP When:

Criterion	Example
✅ Performance critical	Sensor fusion at 1kHz, control loops at 10kHz
✅ Type known at compile-time	Different sensor types (Lidar, Radar, Camera)
✅ Policy-based design	Kalman filter variants, path planning algorithms
✅ Mixin behaviors	Add logging, caching, metrics without overhead
✅ Static interface enforcement	Compile-time API contract checking

📌 3. CRTP vs Virtual Polymorphism Comparison (3/4)

Use Virtual Functions When:

Criterion	Example
✅ True runtime polymorphism	Plugin system for sensors/algorithms
✅ Type unknown at compile-time	Loading algorithm from config file
✅ Heterogeneous containers	`std::vector<std::unique_ptr<Sensor>>`
✅ Simplicity over performance	Non-critical data processing
✅ Dynamic behavior changes	Switching algorithms at runtime

Hybrid Approach:

```
// Use both for different purposes
class ISensor {  // Virtual for runtime polymorphism
    virtual ~ISensor() = default;
    virtual void initialize() = 0;
};

template <typename Derived>
class SensorBase : public ISensor {  // CRTP for performance
    void processFrame() {
        static_cast<Derived*>(this)->fastProcessing();  // Inlined
    }
};

class LidarSensor : public SensorBase<LidarSensor> {
    void initialize() override { /* runtime setup */ }
    void fastProcessing() { /* compile-time optimized */ }
};
```

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Theory

📌 3. CRTP vs Virtual Polymorphism Comparison (4/4)

Decision Matrix:

```
Need runtime type selection? ──YES──> Virtual Functions
         │
         NO
         ↓
Performance critical (>100Hz)? ──YES──> CRTP
         │
         NO
         ↓
Complex behavior composition? ──YES──> CRTP (mixins)
         │
         NO
         ↓
Simple static polymorphism? ──YES──> CRTP
```

---

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Edge Cases

📌 1. Name Hiding in Template Base Classes (1/2)

Problem: Unqualified member access in derived classes doesn't find members of dependent template base classes due to two-phase lookup.

```
// ❌ Name hiding issue
template <typename T>
class Base {
public:
    void helper() { std::cout << "Base helper\n"; }
};

template <typename T>
class Derived : public Base<T> {
public:
    void method() {
        helper();  // ❌ Error: 'helper' not found!
        // Two-phase lookup doesn't consider dependent base
    }
};

// ✅ Fix 1: Use this->
template <typename T>
class DerivedFixed1 : public Base<T> {
public:
    void method() {
        this->helper();  // ✅ Works - qualified lookup
    }
};

// ✅ Fix 2: Using declaration
template <typename T>
class DerivedFixed2 : public Base<T> {
    using Base<T>::helper;  // ✅ Bring name into scope
public:
    void method() {
        helper();  // ✅ Works now
    }
};

// ✅ Fix 3: Explicit qualification
// ...continues on next slide
```

📌 1. Name Hiding in Template Base Classes (2/2)

[Code continues across slides]

```
// ...continued from previous slide
template <typename T>
class DerivedFixed3 : public Base<T> {
public:
    void method() {
        Base<T>::helper();  // ✅ Works - fully qualified
    }
};
```

Key Takeaway: Always use `this->` or explicit qualification when accessing base class members in template-dependent contexts.

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Edge Cases

📌 2. Infinite Recursion Without Implementation Override

Problem: Forgetting to implement required methods in derived class causes infinite recursion or compilation failure.

```
// ❌ Infinite recursion
template <typename T>
class Base {
public:
    void interface() {
        static_cast<T*>(this)->interface();  // ❌ Calls itself infinitely!
    }
};

class Derived : public Base<Derived> {
    // ❌ Missing implementation - causes infinite recursion
};

// ✅ Correct pattern
template <typename T>
class BaseCorrect {
public:
    void interface() {
        static_cast<T*>(this)->interfaceImpl();  // ✅ Calls different method
    }
};

class DerivedCorrect : public BaseCorrect<DerivedCorrect> {
public:
    void interfaceImpl() {  // ✅ Implementation provided
        std::cout << "Implementation\n";
    }
};
```

Key Takeaway: Use different method names (e.g., `interface()` in base calls `interfaceImpl()` in derived) to avoid infinite recursion.

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Edge Cases

📌 3. Multiple CRTP Inheritance Ambiguity (1/2)

Problem: Inheriting from multiple CRTP bases can cause ambiguous `this` pointer casts.

```
// ❌ Potential ambiguity
template <typename T>
class Logger {
public:
    void log(const std::string& msg) {
        std::cout << "[" << static_cast<T*>(this)->name() << "] " << msg << "\n";
    }
};

template <typename T>
class Cacheable {
    mutable std::optional<int> cache;
public:
    int compute() {
        if (!cache) {
            cache = static_cast<T*>(this)->computeImpl();  // ✅ OK - different method
        }
        return *cache;
    }
};

// ✅ Multiple inheritance works if methods don't conflict
class MyClass : public Logger<MyClass>, public Cacheable<MyClass> {
public:
    std::string name() const { return "MyClass"; }
    int computeImpl() { return 42; }
};

// ❌ Ambiguous base if both bases have same method
template <typename T>
class PolicyA {
public:
    void execute() {
        static_cast<T*>(this)->impl();
    }
};

template <typename T>
// ...continues on next slide
```

📌 3. Multiple CRTP Inheritance Ambiguity (2/2)

[Code continues across slides]

```
// ...continued from previous slide
class PolicyB {
public:
    void execute() {  // ❌ Same name as PolicyA::execute()
        static_cast<T*>(this)->impl();
    }
};

class Ambiguous : public PolicyA<Ambiguous>, public PolicyB<Ambiguous> {
public:
    void callExecute() {
        // execute();  // ❌ Error: ambiguous!
        PolicyA<Ambiguous>::execute();  // ✅ Qualify explicitly
    }
    void impl() { }
};
```

Key Takeaway: When using multiple CRTP inheritance, ensure base class interface methods have distinct names or use explicit qualification.

📌 4. Incomplete Type Issues with sizeof()

Problem: The derived class is incomplete when the base class template is instantiated, causing issues with `sizeof` and other type traits.

```
// ❌ Incomplete type issue
template <typename T>
class Base {
    char buffer[sizeof(T)];  // ❌ Error: sizeof incomplete type 'T'
public:
    Base() {
        std::cout << "Size: " << sizeof(T) << "\n";  // ❌ Error
    }
};

class Derived : public Base<Derived> {  // Derived is incomplete here
    int data;
};

// ✅ Fix: Delay sizeof usage until derived is complete
template <typename T>
class BaseFixed {
public:
    void printSize() {
        std::cout << "Size: " << sizeof(T) << "\n";  // ✅ OK - called after construction
    }
};

class DerivedFixed : public BaseFixed<DerivedFixed> {
    int data;
};

DerivedFixed obj;
obj.printSize();  // ✅ Works - Derived is complete now
```

Key Takeaway: Avoid using `sizeof(Derived)` or type traits in base class constructors or as non-type template parameters. Use them in member functions called after object construction.

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Edge Cases

📌 5. Slicing with Value Semantics (1/2)

Problem: Assigning/copying through base class reference causes slicing since CRTP uses static dispatch.

```
template <typename T>
class Base {
public:
    void interface() {
        static_cast<T*>(this)->impl();
    }
};

class Derived1 : public Base<Derived1> {
public:
    int data = 1;
    void impl() { std::cout << "Derived1\n"; }
};

class Derived2 : public Base<Derived2> {
public:
    int data = 2;
    void impl() { std::cout << "Derived2\n"; }
};

// ❌ CRTP doesn't provide runtime polymorphism
void process(Base<Derived1>& obj) {  // ❌ Can only accept Derived1, not Derived2
    obj.interface();
}

// ❌ Cannot have vector of mixed CRTP types
// std::vector<Base<???>> vec;  // ❌ What template parameter?

// ✅ Use template function for generic processing
template <typename T>
void processGeneric(Base<T>& obj) {
    obj.interface();
}

Derived1 d1;
Derived2 d2;
processGeneric(d1);  // ✅ Works
processGeneric(d2);  // ✅ Works
```

📌 5. Slicing with Value Semantics (2/2)

Key Takeaway: CRTP does NOT provide runtime polymorphism. Use template functions, not base class pointers/references, when you need to process different derived types generically.

---

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Code Examples

📌 1. Basic CRTP - Static Polymorphism (Easy) (1/2)

Key Concepts:
- CRTP base template `Printer<Derived>`
- Static downcast with `static_cast<Derived*>(this)`
- Compile-time polymorphism
- Generic functions using template parameter matching

```
#include <iostream>
#include <string>

// CRTP base class
template <typename Derived>
class Printer {
public:
    void print() const {
        // Static downcast to derived type
        static_cast<const Derived*>(this)->printImpl();
    }
};

// Derived classes
class IntPrinter : public Printer<IntPrinter> {
    int value;
public:
    IntPrinter(int v) : value(v) {}

    void printImpl() const {
        std::cout << "Integer: " << value << "\n";
    }
};

class StringPrinter : public Printer<StringPrinter> {
    std::string value;
public:
    StringPrinter(const std::string& v) : value(v) {}

    void printImpl() const {
        std::cout << "String: " << value << "\n";
    }
// ...continues on next slide
```

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Code Examples

📌 1. Basic CRTP - Static Polymorphism (Easy) (2/2)

[Code continues across slides]

```
// ...continued from previous slide
};

// Generic function using CRTP
template <typename T>
void genericPrint(const Printer<T>& printer) {
    printer.print();  // Resolved at compile-time
}

int main() {
    IntPrinter ip(42);
    StringPrinter sp("Hello");

    ip.print();  // Calls IntPrinter::printImpl()
    sp.print();  // Calls StringPrinter::printImpl()

    genericPrint(ip);  // Template function
    genericPrint(sp);

    return 0;
}

// Output:
// Integer: 42
// String: Hello
// Integer: 42
// String: Hello
```

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Code Examples

📌 2. Shape Hierarchy with CRTP (Mid) (1/3)

Key Concepts:
- Reusable shape interface via CRTP
- `display()` method in base calls derived methods
- Type-specific additional methods (getRadius(), getDimensions())
- Zero virtual function overhead

```
#include <iostream>
#include <cmath>
#include <utility>

// CRTP base for shapes
template <typename Derived>
class Shape {
public:
    double area() const {
        return static_cast<const Derived*>(this)->areaImpl();
    }

    void display() const {
        std::cout << static_cast<const Derived*>(this)->name()
                  << " - Area: " << area() << "\n";
    }
};

// Circle
class Circle : public Shape<Circle> {
    double radius;
public:
    Circle(double r) : radius(r) {}

    double areaImpl() const {
        return M_PI * radius * radius;
    }

    std::string name() const {
        return "Circle";
    }

// ...continues on next slide
```

📌 2. Shape Hierarchy with CRTP (Mid) (2/3)

```
// ...continued from previous slide
    double getRadius() const { return radius; }
};

// Rectangle
class Rectangle : public Shape<Rectangle> {
    double width, height;
public:
    Rectangle(double w, double h) : width(w), height(h) {}

    double areaImpl() const {
        return width * height;
    }

    std::string name() const {
        return "Rectangle";
    }

    std::pair<double, double> getDimensions() const {
        return {width, height};
    }
};

// Triangle
class Triangle : public Shape<Triangle> {
    double base, height;
public:
    Triangle(double b, double h) : base(b), height(h) {}

    double areaImpl() const {
        return 0.5 * base * height;
    }

    std::string name() const {
        return "Triangle";
    }
};

// Generic function to display any shape
template <typename T>
void showShape(const Shape<T>& shape) {
// ...continues on next slide
```

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Code Examples

📌 2. Shape Hierarchy with CRTP (Mid) (3/3)

[Code continues across slides]

```
// ...continued from previous slide
    shape.display();
}

int main() {
    Circle circle(5.0);
    Rectangle rectangle(4.0, 6.0);
    Triangle triangle(3.0, 4.0);

    std::cout << "Shape Areas:\n";
    circle.display();
    rectangle.display();
    triangle.display();

    std::cout << "\nUsing generic function:\n";
    showShape(circle);
    showShape(rectangle);
    showShape(triangle);

    return 0;
}

// Output:
// Shape Areas:
// Circle - Area: 78.5398
// Rectangle - Area: 24
// Triangle - Area: 6
//
// Using generic function:
// Circle - Area: 78.5398
// Rectangle - Area: 24
// Triangle - Area: 6
```

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Code Examples

📌 3. Logging Policy with CRTP (Advanced) (1/3)

Key Concepts:
- Logging policy as CRTP mixin
- Automatic timestamped logging for derived classes
- `this->` required for template-dependent name lookup
- Each derived class provides `name()` for context

```
#include <iostream>
#include <string>
#include <chrono>
#include <iomanip>

// Logger policy using CRTP
template <typename Derived>
class Logger {
public:
    void log(const std::string& message) const {
        auto now = std::chrono::system_clock::now();
        auto time = std::chrono::system_clock::to_time_t(now);

        std::cout << "[" << std::put_time(std::localtime(&time), "%H:%M:%S")
                  << "] [" << getDerivedName() << "] "
                  << message << "\n";
    }

protected:
    std::string getDerivedName() const {
        return static_cast<const Derived*>(this)->name();
    }
};

// Sensor class with logging
class LidarSensor : public Logger<LidarSensor> {
    double maxRange;
    int pointCount;

public:
    LidarSensor(double range, int points)
        : maxRange(range), pointCount(points) {
// ...continues on next slide
```

📌 3. Logging Policy with CRTP (Advanced) (2/3)

```
// ...continued from previous slide
        this->log("LidarSensor initialized");
    }

    void scan() {
        this->log("Starting scan...");
        // Simulate scan
        this->log("Scan complete - " + std::to_string(pointCount) + " points");
    }

    std::string name() const { return "LiDAR"; }
};

class RadarSensor : public Logger<RadarSensor> {
    double frequency;

public:
    RadarSensor(double freq) : frequency(freq) {
        this->log("RadarSensor initialized");
    }

    void detect() {
        this->log("Detecting objects...");
        // Simulate detection
        this->log("Detection complete");
    }

    std::string name() const { return "Radar"; }
};

int main() {
    LidarSensor lidar(100.0, 256000);
    lidar.scan();

    std::cout << "\n";

    RadarSensor radar(77.5);
    radar.detect();

    return 0;
}
// ...continues on next slide
```

📌 3. Logging Policy with CRTP (Advanced) (3/3)

[Code continues across slides]

```
// ...continued from previous slide

// Output (sample):
// [14:30:15] [LiDAR] LidarSensor initialized
// [14:30:15] [LiDAR] Starting scan...
// [14:30:15] [LiDAR] Scan complete - 256000 points
//
// [14:30:15] [Radar] RadarSensor initialized
// [14:30:15] [Radar] Detecting objects...
// [14:30:15] [Radar] Detection complete
```

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Code Examples

📌 4. Caching Policy with CRTP (Advanced) (1/3)

Key Concepts:
- Caching policy as CRTP mixin
- `mutable std::optional` for cache storage
- Cache invalidation on state changes
- Transparent caching without modifying core logic

```
#include <iostream>
#include <optional>
#include <cmath>

// Caching policy using CRTP
template <typename Derived>
class Cacheable {
    mutable std::optional<double> cache;

public:
    double compute() const {
        if (!cache) {
            std::cout << "[Cache Miss] Computing...\n";
            cache = static_cast<const Derived*>(this)->computeImpl();
        } else {
            std::cout << "[Cache Hit] Returning cached value\n";
        }
        return *cache;
    }

    void invalidateCache() {
        cache.reset();
    }
};

// Expensive computation class with caching
class ExpensiveOperation : public Cacheable<ExpensiveOperation> {
    double radius;

public:
    ExpensiveOperation(double r) : radius(r) {}

// ...continues on next slide
```

📌 4. Caching Policy with CRTP (Advanced) (2/3)

```
// ...continued from previous slide
    double computeImpl() const {
        // Simulate expensive computation
        double result = 0.0;
        for (int i = 0; i < 1000000; i++) {
            result += std::sin(radius) * std::cos(radius);
        }
        return result;
    }

    void setRadius(double r) {
        radius = r;
        invalidateCache();  // Cache no longer valid
    }

    double getRadius() const { return radius; }
};

int main() {
    ExpensiveOperation op(3.14);

    // First call - cache miss
    double result1 = op.compute();
    std::cout << "Result: " << result1 << "\n\n";

    // Second call - cache hit
    double result2 = op.compute();
    std::cout << "Result: " << result2 << "\n\n";

    // Change data - invalidate cache
    op.setRadius(6.28);

    // Third call - cache miss again
    double result3 = op.compute();
    std::cout << "Result: " << result3 << "\n";

    return 0;
}

// Output:
// [Cache Miss] Computing...
// ...continues on next slide
```

📌 4. Caching Policy with CRTP (Advanced) (3/3)

[Code continues across slides]

```
// ...continued from previous slide
// Result: 2.50457e+06
//
// [Cache Hit] Returning cached value
// Result: 2.50457e+06
//
// [Cache Miss] Computing...
// Result: -5.00885e+05
```

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Code Examples

📌 5. Multiple Policy Composition (Advanced) (1/4)

Key Concepts:
- Multiple CRTP inheritance for policy composition
- Policies interact (Logger used by Cacheable)
- Zero runtime overhead
- Compile-time composition of behaviors

```
#include <iostream>
#include <string>
#include <optional>

// Logger policy
template <typename Derived>
class Logger {
public:
    void log(const std::string& msg) const {
        std::cout << "[" << static_cast<const Derived*>(this)->name()
                  << "] " << msg << "\n";
    }
};

// Caching policy
template <typename Derived>
class Cacheable {
    mutable std::optional<double> cache;

public:
    double getOrCompute() const {
        if (!cache) {
            static_cast<const Derived*>(this)->log("Computing value...");
            cache = static_cast<const Derived*>(this)->computeImpl();
            static_cast<const Derived*>(this)->log("Value cached");
        } else {
            static_cast<const Derived*>(this)->log("Using cached value");
        }
        return *cache;
    }

    void invalidateCache() {
// ...continues on next slide
```

📌 5. Multiple Policy Composition (Advanced) (2/4)

```
// ...continued from previous slide
        cache.reset();
        static_cast<const Derived*>(this)->log("Cache invalidated");
    }
};

// Metrics policy
template <typename Derived>
class Metricsable {
    mutable int callCount = 0;

public:
    void incrementCalls() const {
        callCount++;
    }

    int getCallCount() const { return callCount; }
};

// Class using multiple policies
class SmartCalculator
    : public Logger<SmartCalculator>,
      public Cacheable<SmartCalculator>,
      public Metricsable<SmartCalculator> {

    double value;

public:
    SmartCalculator(double v) : value(v) {
        this->log("SmartCalculator initialized");
    }

    double calculate() const {
        this->incrementCalls();
        return this->getOrCompute();
    }

    double computeImpl() const {
        // Expensive computation
        double result = value * value + value;
        return result;
// ...continues on next slide
```

📌 5. Multiple Policy Composition (Advanced) (3/4)

```
// ...continued from previous slide
    }

    void setValue(double v) {
        value = v;
        this->invalidateCache();
    }

    std::string name() const { return "SmartCalculator"; }

    void printStats() const {
        this->log("Total calls: " + std::to_string(this->getCallCount()));
    }
};

int main() {
    SmartCalculator calc(5.0);

    std::cout << "\nFirst calculation:\n";
    std::cout << "Result: " << calc.calculate() << "\n";

    std::cout << "\nSecond calculation (should use cache):\n";
    std::cout << "Result: " << calc.calculate() << "\n";

    std::cout << "\nChanging value:\n";
    calc.setValue(10.0);

    std::cout << "\nThird calculation (cache miss):\n";
    std::cout << "Result: " << calc.calculate() << "\n";

    std::cout << "\nStatistics:\n";
    calc.printStats();

    return 0;
}

// Output:
// [SmartCalculator] SmartCalculator initialized
//
// First calculation:
// [SmartCalculator] Computing value...
// ...continues on next slide
```

📌 5. Multiple Policy Composition (Advanced) (4/4)

[Code continues across slides]

```
// ...continued from previous slide
// [SmartCalculator] Value cached
// Result: 30
//
// Second calculation (should use cache):
// [SmartCalculator] Using cached value
// Result: 30
//
// Changing value:
// [SmartCalculator] Cache invalidated
//
// Third calculation (cache miss):
// [SmartCalculator] Computing value...
// [SmartCalculator] Value cached
// Result: 110
//
// Statistics:
// [SmartCalculator] Total calls: 3
```

What You Learned

Key Takeaways

- ✓ CRTP Pattern Overview
- ✓ CRTP Benefits in Autonomous Vehicles
- ✓ CRTP vs Virtual Polymorphism Comparison
- ✓ Name Hiding in Template Base Classes

→ *Swipe to learn* →